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Uncertainty in Time Series Foundation Models



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Abstract

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Contents

Abstract ii

1 Introduction 1
- 1.1 Motivation . . . 1
- 1.2 Objectives and Approach . . . 1
- 1.3 Outline . . . 1

2 Background 2
- 2.1 Definition and Notation . . . 2
- 2.2 Related Work . . . 3
  - 2.2.1 Classical Statistical Forecasting . . . 3
  - 2.2.2 Deep Learning and Global Models . . . 4
  - 2.2.3 Transformers . . . 4
  - 2.2.4 Patching . . . 4
  - 2.2.5 Probabilistic Forecasting . . . 5
  - 2.2.6 Time Series Foundation Models . . . 5
  - 2.2.7 Multivariate Foundation Models with covariates . . . 6
  - 2.2.8 Efficiency and Streaming . . . 6
  - 2.2.9 Benchmarks . . . 6

3 Time Series Foundation Models 7
- 3.1 Chronos-2 . . . 7
- 3.2 TiRex-2 . . . 8
- 3.3 Timestral . . . 9

4 Experiments 10

5 Results 11

6 Conclusion and Outlook 12

Bibliography 13

Appendix A An Appendix 16

Chapter 1

Introduction

1.1 Motivation

Space for your introduction. A typical introduction consists of a motivation for your thesis, your objectives, your research approach, and finishes with an outlook over your work. Of course, different structuring and naming of sections is possible.

1.2 Objectives and Approach

1.3 Outline

Chapter 2

Background

2.1 Definition and Notation

What is time series forecasting? Given a known sequence of values $X_{1:T} = [x_1, \ldots, x_T]$ with $x_t \in \mathbb{R}$, the goal of time series forecasting is to predict the next H future time steps $X_{T+1:T+H} = [x_{T+1}, \ldots, x_{T+H}]$, where T denotes the context length and H the forecast horizon. This setting describes the univariate case.

To make the separation between known context and future targets more clear, throughout this thesis the future targets are denoted as $Y_{1:H} = [y_1, \ldots, y_H]$ with $y_h \in \mathbb{R}$ and the final model output is denoted as $\hat{Y}_{1:H} = [\hat{y}_1, \ldots, \hat{y}_H]$ with $\hat{y}_h \in \mathbb{R}$.

Modern time series models such as Chronos-2 (Ansari et al. [2]), Toto 2.0 (Khwaja et al. [16]) and TiRex-2 (Podest et al. [21]) extend the univariate case to multivariate and incorporate additional information by using one or multiple covariates for the historic context and forecast horizon. Following the notation of Podest et al. [21] with minor adaptations, the input now consists of the target series $X^{tgt}_{1:T} \in \mathbb{R}^{V^{tgt} \times T}$, past covariates $X^{pcov}_{1:T} \in \mathbb{R}^{V^{pcov} \times T}$ which are observed up to time T, and future covariates $X^{fcov}_{1:T+H} \in \mathbb{R}^{V^{fcov} \times (T+H)}$ which are also known over the forecast horizon. The total number of variates is $V = V^{tgt} + V^{pcov} + V^{fcov}$. The predictions $Y_{1:H} \in \mathbb{R}^{V^{tgt} \times H}$ also gain an additional variate dimension.

While classic approaches such as ARIMA (Box and Jenkins [7]) primarily produce point forecasts, recent models learn to approximate the conditional distribution over future outcomes:

$$\mathcal{P}\left(Y_{1:H} | X^{tgt}_{1:T}, X^{pcov}_{1:T}, X^{fcov}_{1:T+H}\right) \tag{2.1}$$

This distribution is often approximated by K quantile levels $\mathcal{Q} = \{\tau_1, \ldots, \tau_K\} \subset (0, 1)$ learned by a parametrised function g_θ:

$$\left\{\mathcal{Q}_\tau\left(Y_{1:H} | X^{tgt}_{1:T}, X^{pcov}_{1:T}, X^{fcov}_{1:T+H}\right)\right\}_{\tau \in \mathcal{Q}} \approx g_\theta\left(X^{tgt}_{1:T}, X^{pcov}_{1:T}, X^{fcov}_{1:T+H}\right) \tag{2.2}$$

where $\mathcal{Q}_\tau(\cdot|\cdot)$ defines the conditional τ-quantile.

Further, in recent models an input sequence is often splitted to non-overlapping patches of length P. After extending all series over the forecast horizon, this transformation reshapes a time series $X \in \mathbb{R}^{V \times T}$ into $X \in \mathbb{R}^{V \times N \times P}$, where $N = \lceil (T+H)/P \rceil$ denotes the number of patches. This applies to targets, past covariates and future covariates. Usually, this patches are further embedded into a higher dimension D to serve as input to the model. As a result of patching and embedding the model input has the shape $Z \in \mathbb{R}^{V \times N \times D}$. The final hidden state is then transformed to quantile predictions $\hat{Y} \in \mathbb{R}^{V^{tgt} \times H \times K}$.

To complete the notation, B denotes the batch size. Since this notations describe a general form of modern time series models and of the forecasting process, individual models may deviate slightly from it. Such differences are described in the respective sections.

- X: model input (split into X^{tgt}, X^{pcov}, X^{fcov})
- Y: future targets; $\hat{Y}$: model output
- Z: embedding output; $\hat{Z}$: hidden state or intermediate model output
- T: context length
- H: forecast horizon
- V: number of variates (V^{tgt}, V^{pcov}, V^{fcov})
- P: patch size
- N: number of patches
- K: number of quantiles ($\tau \in \mathcal{Q}$)
- D: model dimension
- B: batch size

2.2 Related Work

This chapter reviews the development of time series forecasting from classical statistical methods to recent foundation models. These approaches can be divided into three main categories. (I) Local models such as ARIMA (Box and Jenkins [7]) and exponential smoothing (Hyndman et al. [15]) fit a separate set of parameters for each time series. (II) Dataset-specific global models such as DeepAR (Salinas et al. [24]) and N-BEATS (Oreshkin et al. [20]) share their parameters across all series of a dataset, but have to be retrained for every new dataset or task. (III) Zero-shot foundation models such as Chronos (Ansari et al. [3]), TiRex (Auer et al. [5]) and Toto (Cohen et al. [9]) follow the paradigm of large language models. They are trained on a big set of data with various time series of different domains and generate forecasts for unseen sequences without requiring additional training or fine-tuning. This chapter also covers design choices which are implemented by this category of models, such as probabilistic forecasting with quantiles, patch-based input representations, learning from additional covariates and efficient streaming.

2.2.1 Classical Statistical Forecasting

Classical forecasting methods model each time series individually. ARIMA (Box and Jenkins [7]) describes a series as a linear combination of its own past values and past forecast errors, after differencing to remove trends. While the classical Box-Jenkins methodology selects the model manually, Auto-ARIMA (Hyndman and Khandakar [14]) automates this selection. Exponential smoothing (Hyndman et al. [15]) decompose a series into error, trend and seasonal components (state space form labeled as ETS) that are updated recursively. The Theta method (Assimakopoulos and Nikolopoulos [4]) decomposes a series into two θ-lines: a linear trend line that is extrapolated directly and a line with amplified local curvature that is forecast with simple exponential smoothing. The final forecast is the average of both.

Although deep learning models often outperform the classical approaches, they remain a strong baseline in current benchmarks. They are easy to interpret and are computationally cheap. But they are limited to single series since each time series needs its own fitting step and no knowledge is transferred between different series.

2.2.2 Deep Learning and Global Models

A key shift was the move from local to global models, which share one set of parameters across multiple time series. Global models learn patterns that are shared across series and can therefore forecast series with little history. One enabler were, recurrent neural networks, specially the LSTM (Hochreiter and Schmidhuber [13]), were among the first deep architectures applied to forecasting. They process a series step by step while maintaining a hidden state. DeepAR (Salinas et al. [24]) trains an autoregressive LSTM on a dataset and predicts the parameters of a likelihood at each step, from which forecasts are obtained by sampling. N-BEATS (Oreshkin et al. [20]) showed that a deep architecture of fully connected layers with residual connections can outperform statistical methods. N-HiTS (Challu et al. [8]) extended this design with multi-rate sampling and hierarchical interpolation to improve long-horizon forecasting.

2.2.3 Transformers

Due to the success of the Transformer (Vaswani et al. [28]) in natural language processing, the architecture was soon adapted to time series forecasting. Unlike recurrent networks, self-attention connects every pair of positions directly, which promises to capture long-range dependencies without the step-by-step propagation of a hidden state. The first adaptations, however, used each time step as one token, typically embedding the values of all variates at that time step together with timestamp features. Since the cost of self-attention grows quadratically with the number of tokens, long contexts and horizons quickly became expensive.

The Temporal Fusion Transformer (Lim et al. [17]) combines recurrence with attention. It expands the input space by adding static covariates, past-observed inputs that are only available up to the forecast origin, and known future inputs that are available over the horizon. The model encode past inputs and decode future inputs via a LSTM layer and adds a temporal self-attention layer. Informer (Zhou et al. [34]) restricted attention to the most informative queries (ProbSparse attention) and replaced autoregressive decoding with a decoder that predicts the whole horizon in a single forward pass, avoiding error accumulation over long horizons. Autoformer (Wu et al. [29]) integrated a decomposition into trend and seasonal components into the architecture. FEDformer (Zhou et al. [35]) moved attention into the frequency domain and operated on a small subset of Fourier modes, reducing the complexity to linear in the sequence length. Their effectiveness was questioned by Zeng et al. [31], who showed that a simple linear model can outperform them on common long-horizon benchmarks. This raised the question of whether attention over single time steps is a suitable representation for time series at all.

2.2.4 Patching

PatchTST (Nie et al. [19]) addressed the issue of attention for single time steps by splitting each series into patches of length P and treating every patch as one token. Patching reduces the sequence length from T to $N \approx T/P$, which lowers the cost of attention, and

gives each token a local context of several time steps that carries more information than a single value. Patching has since become the standard input representation of time series foundation models. In addition, PatchTST processes every variate independently with shared weights (channel independence), which proved more robust than mixing variates naively. But channel independence, ignores information shared between variates. Crossformer (Zhang and Yan [33]) embeds each variate patch-wise and alternates between attention over time and attention over variates.

2.2.5 Probabilistic Forecasting

Point forecasts do not express how uncertain a prediction is, which is essential for decision making and for downstream tasks such as anomaly detection. Quantile regression addresses this by estimating conditional quantiles directly. The τ-quantile is the value that the future observation falls below with probability τ, so for example the 0.1- and 0.9-quantiles together span an 80% prediction interval. For a single target series, the quantile loss over the forecast horizon is

$$\mathcal{L} = \frac{1}{K \cdot H} \sum_{h=1}^{H} \sum_{\tau \in \mathcal{Q}} \begin{cases} \tau \left(y_h - \hat{y}_h^\tau \right) & \text{if } \hat{y}_h^\tau \leq y_h, \\ (1-\tau) \left(\hat{y}_h^\tau - y_h \right) & \text{otherwise,} \end{cases} \tag{2.3}$$

where y_h is the observed value at horizon step h and $\hat{y}_h^\tau$ the predicted τ-quantile. The loss penalises under-prediction with weight τ and over-prediction with weight $1 - \tau$, so that its minimiser is the τ-quantile of the conditional distribution. For $\tau = 0.5$, it reduces to half the absolute error and yields the median.

A model predicts one value per quantile level in a fixed set $\mathcal{Q}$ with $K = |\mathcal{Q}|$ levels, and the choice of this set differs between models. The Temporal Fusion Transformer (Lim et al. [17]) uses $K = 3$ levels $\{0.1, 0.5, 0.9\}$, while TiRex-1 (Auer et al. [5]) and TiRex-2 (Podest et al. [21]) use $K = 9$ levels $\{0.1, 0.2, \ldots, 0.9\}$. Chronos-2 (Ansari et al. [2]) extends this to $K = 21$ levels, $\{0.01, 0.99\}$ together with $\{0.05, 0.10, \ldots, 0.95\}$ in steps of 0.05, to better capture the tails of the distribution and thus rare events.

2.2.6 Time Series Foundation Models

Time series foundation models commonly use the zero-shot forecast paradigm. This means, the model is pretrained on large and diverse collections of real and synthetic series and predict unseen series, without any training or fine-tuning on the target data. TimeGPT (Garza et al. [11]) and Lag-Llama (Rasul et al. [23]) were early examples of this approach. Chronos (Ansari et al. [3]) scales and quantises time series values into a fixed vocabulary and trains a T5 language model with a cross-entropy loss, treating forecasting as next-token prediction. TimesFM (Das et al. [10]) uses a decoder-only Transformer over patches.

TiRex (Auer et al. [5]) breaks apart from the Transformer and builds on the xLSTM architecture (Beck et al. [6]). It mention that the state tracking of recurrent models is beneficial for forecasting and introduces contiguous patch masking during training to improve long-horizon predictions. Toto (Cohen et al. [9]) is a decoder-only model trained with a strong focus on observability data and uses a Student-T mixture model head instead of the common quantile head.

Most of these first-generation foundation models are univariate: each series is forecast in isolation, and covariates cannot be used or only through additional mechanisms.

2.2.7 Multivariate Foundation Models with covariates

The recent generation of foundation models supports multivariate forecasting and covariates natively. The Chronos-2 (Ansari et al. [2]) transformer stack adds a time attention layer which aggregates information across patches within a single time series and additionally a group attention layer which aggregates information across all series within a group at each patch. TiRex-2 (Podest et al. [21]) combines a bidirectional xLSTM time mixer with attention over variates at each patch position. The forward recurrence processes all variates, while a reverse recurrence runs only over the known future covariates, so that information about the forecast horizon can flow backwards into the context.

These models follow the general structure introduced in Section 2.1: patched inputs with targets, past covariates and future covariates stacked along the variate axis, alternating mixing along the time and variate axes, and a quantile output head. They differ, however, in the type of time mixer, in how cross-variate information is exchanged, and in how future covariates are processed.

2.2.8 Efficiency and Streaming

While Transformers dominate sequence modelling, their attention cost grows quadratically with the sequence length, and autoregressive inference requires caching keys and values for the whole context. This has renewed interest in architectures with a constant-size state.

xLSTM (Beck et al. [6]) revisits the LSTM with exponential gating and modified memory structures. Its sLSTM variant keeps a scalar memory with recurrent connections between hidden states, which enables state tracking but requires sequential computation. The mLSTM variant replaces the scalar memory with a matrix memory, so it can be trained in parallel like linear attention. Both variants can be evaluated step by step, so that processing a new input costs $\mathcal{O}(1)$ with respect to the length of the context. TiRex and TiRex-2 build on these cells.

In applications, such as monitoring, observations arrive continuously and forecasts must be updated whenever new data becomes available. For unidirectional recurrent models, streaming inference is straightforward, since the state after the last observation summarises the whole context and only the new inputs have to be processed. For a Transformer without caching, every update requires recomputing the forward pass over the full context. This makes models, such as Tirex-2 (Podest et al. [21]) a better fit for this applications.

2.2.9 Benchmarks

Zero-shot forecasting models are evaluated on collections of datasets that were not used for training. Common benchmarks in this field are GIFT-Eval (Aksu et al. [1]) and fev-bench (Shchur et al. [26]). fev-bench in particular includes tasks with covariates.

A complementary approach uses synthetic data from known dynamical systems. dysts (Gilpin [12]) is a collection of over 130 chaotic systems with published parameters, together with tools to generate trajectories of arbitrary length. Zhang and Gilpin [32] used dysts to evaluate a time series foundation model zero-shot on chaotic systems. Besides pointwise error, they measured how long forecasts stay accurate.

Chapter 3

Time Series Foundation Models

Since it was not feasible to test all current available time series foundation models, a selection had to be made. The choice was fallen on three recent models. Chronos-2 (Ansari et al. [2]), TiRex-2 (Podest et al. [21]) and Timestral (**timestral** [**timestral**]). All three models support multivariate forecasting with additional past-known X^{pcov} and future-known X^{fcov} covariates. They also use patching for input processing and return quantile predictions via a quantile head. Each model has its unique feature: Chronos-2 is a encoder-only transformer that follows the design of the T5 encoder (Raffel et al. [22]); TiRex-2 builds on xLSTM (Beck et al. [6]) and allows for efficient streaming with constant memory; Timestral is the most recent transformer that currently beats every other foundation model on common benchmarks and is mostly trained on synthetic data. A more detailed introduction to the three models is made in the following subchapters.

3.1 Chronos-2

Chronos-2 extend its predecessor to multivariate forecasting and allows past-known and future-known covariates as additional inputs. It adds many new techniques the first implementation was lacking such as patching, group attention and quantile predictions.

The input series is split into past-known values $X_p \in \mathbb{R}^{V \times T}$ and future-known values $X_f \in \mathbb{R}^{V \times H}$ where targets and past-known values are set to missing values. Categorical covariates are encoded either by target encoding (Micci-Barreca [18]) for univariate targets or ordinal encoding for multivariate targets.

Since input values can have any scale, all input values are scaled by standardization with an additional arcsinh transformation step, which is commonly used in recent foundation models. The formula can be written as:

$$\tilde{x} = \operatorname{arcsinh}\left(\frac{x - \mu_p}{\sigma_p}\right) \tag{3.1}$$

where μ_p is the mean and σ_p the variance of past-known values. Missing values are excluded from computing μ_p and σ_p. The additional transformation helps to reduce the influence of outliers on the objective function. The previous iteration of Chronos used every single time step as input to the model. This changes in Chronos-2. It applies patching to the input values X_p and X_f with zero padding if T and H are not multiplies of P. They also concatenate two additional meta features. (I) A time index which encodes the relative position of each time step. (II) A mask for missing values. The stack of patches is then embedded via a residual network $f_\phi^{\text{in}} : \mathbb{R}^{3P} \to \mathbb{R}^D$, where ϕ denotes the model parameters. Between past and future embeddings a special *REG* token is inserted which serves as separator and attention sink (Xiao et al. [30]). The full embedded input is then feed into a transformer stack.

The transformer stack only uses an encoder and follows the design of the T5 encoder (Raffel et al. [22]). The original published model has 12 layers where outputs of on layer are feed back to the next layer as input and consists of three parts: Time Attention, Group Attention and a Feed-Forward Network. The time attention layer applies self-attention on the time axis. It uses rotary position embeddings (RoPE) (Su et al. [27]) for the relative position embeddings. The group attention layer enables learning from other time series and makes use of future and past covariates. It aggregates information across time series that belong to the same group for a given patch. The group IDs given to each time series inside a batch depends on the forecast task. For univariate forecasting each time series gets a unique id. For multivariate forecasting each time series share the same group id. And for forecasting with covariates, each series also share the same group id. Based on this group IDs a group mask is created which makes sure aggregation happens only within the same group. After time attention and group attention the input is passed through a feed-forward network.

Finally, the outputs of the transformer stack are then passed through a residual network that produces the quantile forecast $\hat{Y} \in \mathbb{R}^{H \times V^{tgt} \times K}$ and then rescaled back to the original scale.

3.2 TiRex-2

As Chronos-2, TiRex-2 extend its predecessor to multivariate forecasting and allows past-known and future-known covariates as additional inputs. While TiRex only uses sLSTM layers from the xLSTM architecture, TiRex-2 uses both sLSTM and mLSTM layers and uses synthetic multivariate data for training which is generated by synthetic multivariate coupling.

The extended inputs $X \in \mathbb{R}^{V \times (T+H)}$ for TiRex-2 are preprocessed in three steps: scaling, patching and embedding. The predecessor used standard z-score normalization. TiRex-2 uses the same scaling technique as in Chronos-2, a z-score normalization followed by an arcsinh transform that compresses heavy tails.

$$\tilde{x} = (1 - b) \operatorname{arcsinh}\left(\frac{x - \mu_p}{\sigma_p}\right) + bx \tag{3.2}$$

It extends this scaling with a binary gate b that bypasses the transformation for variates taking only the values $\{0, 1\}$. Without this bypass, the standardized value of the positive level would grow as $1/\sqrt{\bar{p}}$ with decreasing frequency $\bar{p}$ of ones, so the same event would be encoded differently depending on how sparse it is in the context window. After scaling, each variate is padded and patched into $N = \lceil (T + H)/P \rceil$ non overlapping patches of length P and then embedded to a higher dimension D via a two-layer residual MLP $f_\phi^{\text{in}} : \mathbb{R}^P \to \mathbb{R}^D$ where ϕ denotes the model parameters. The input tensor $Z \in \mathbb{R}^{V \times N \times D}$ is then feed into the first block.

The block is build from two parts. The time mixer which process each variate independently along the time axis N. It processes the time series by a bi-directional weight-tied xLSTM (Schmidinger et al. [25]). Target and past-covariate tokens are processed strictly forward, future-known covariates tokens are processed reverse and then combined by a linear fusion layer. The xLSTM alternates between mLSTM and sLSTM blocks. At the end of of the time mixer a MLP with a residual connection is added. The variate mixer uses multi-head self-attention with an asymmetric mask for each group and is inactive for univariate forecasting. The mask allows target queries to read from covariate keys

but covariate queries cannot read target keys. A MLP with residual connection is also added at the end of the variate mixer.

After alternating between time mixer and variate mixer the quantile predictions are then created by a two-layer MLP and scaled back.

Since TiRex-2 uses a recurrent model as a time mixer, it is very efficient on long context. It scales with $\mathcal{O}(V\ N)$ on the time axis, while an attention-based time mixer would scale with $\mathcal{O}(V\ N^2)$. This allows TiRex-2 to be efficient in streaming tasks.

3.3 Timestral

Chapter 4

Experiments

Space for your experiments

Chapter 5

Results

Space for Results

Chapter 6

Conclusion and Outlook

Space for your summary, central conclusions, and an outlook on potential future work.

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Appendix A

An Appendix

Space for an appendix. You can have more than one appendix chapter. Appendices are, of course, optional.